

# Maximillian Railton

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## Technical Skills

**Programming Languages:** Python, C++, C, Java, JavaScript, MATLAB, VHDL, Bash, SQL, HTML/CSS  
**AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras, OpenCV, NumPy, Pandas, Matplotlib, Jupyter, OpenAI Gym  
**ML Specializations:** Computer Vision (Mask R-CNN, CNNs, image segmentation, object detection) • NLP (TF-IDF, mutual information feature selection) • Reinforcement Learning (Q-Learning, Double Q-Learning, Expected SARSA, REINFORCE, Actor-Critic) • Classical ML (KNN, Decision Trees, Logistic Regression, MLPs)  
**Tools & Infrastructure:** Git, Docker, Linux, InfluxDB/Flux, Tox, Flake8, SCPI/PyVISA, GitHub  
**Operating Systems & Hardware:** Linux (Ubuntu, Debian), Windows, macOS • Embedded systems, ASICs, oscilloscopes, signal generators, multimeters, network analyzers

## Education

**McGill University** 2020 – 2026  
*B.Eng. in Computer Engineering • Minor in Applied Artificial Intelligence*  
• Coursework: Reinforcement Learning, Applied Machine Learning, Neural Networks, Computer Vision, Applied AI, Software Testing, Data Structures & Algorithms, Embedded Systems  
• Varsity Track and Field  
**The Country Day School** 2008 – 2020  
*Highschool Diploma*

## Relevant Experience

**HaiLa Technologies** Montreal, QC 2023 – Present  
*Software and Hardware Engineer*  
• Built and maintained a Python instrumentation library (SCPI/PyVISA) for automated RF IoT chip testing, integrating drivers for oscilloscopes, network analyzers, signal generators, and multimeters.  
• Engineered a test automation UI for chip benchmarking; produced a client QA guide; built and ran coverage tests.  
• Wrote InfluxDB Flux queries to ingest time-series data and build dashboards; performed precision hardware modifications and validated chips against design specifications.  
**Nextdriv** Montreal, QC 2026  
*AI Strategy & Business Development*  
• Mapped the Quebec AI EV fleet market, profiled prospect clients across transit and municipal sectors, and synthesized competitive intelligence on AI-driven fleet optimization for executive stakeholder meetings.  
**2Nite App** Montreal, QC 2023 – Present  
*Chief Technology Officer (CTO)*  
• Led full-stack architecture of a mobile event-discovery platform (ticketing, RSVPs, listings) from prototype to production; triaged bugs and coordinated feature rollouts.

## Projects

**Renewable Energy Site Detection via Satellite Imagery**  
*Capstone | Mask R-CNN • PyTorch • Computer Vision • Image Segmentation*  
• Trained Mask R-CNN on satellite imagery to identify non-invasive solar/wind installation sites; implemented full instance segmentation pipeline with preprocessing, anchor generation, and post-processing.  
**AI Resume Parser • Sign Language Classification • Tumor Malignancy Classifier**  
*Python • PyTorch • MLP • CNN • KNN • Decision Trees • Scikit-learn • NLP*  
• Resume parser: automated candidate screening using ML classification to rank resumes against job requirements.  
• Sign language classification: benchmarked MLP architectures vs. ConvNet on MNIST-format hand sign dataset; analyzed depth, regularization, and activation impact on accuracy.  
• Tumor classifier: binary malignancy prediction using a custom decision tree and KNN; evaluated across precision, recall, and AUC with hyperparameter tuning.  
**20 Newsgroups Text Classification • RL Bias & Policy Gradient Study**  
*TF-IDF • Mutual Info • Logistic Regression • Q-Learning • SARSA • REINFORCE • Actor-Critic*  
• Newsgroups: Jupyter pipeline with TF-IDF, mutual information feature selection, and custom logistic regression, decision tree, and KNN implementations with full evaluation metrics.  
• RL study: implemented and compared Q-Learning, Double Q-Learning, Expected SARSA, and policy gradient methods; studied maximization bias and variance reduction across on/off-policy algorithms.

## Additional

**Language Skills:** Native Language: English • Conversational: French (12+ years)  
**Interests:** AI/ML research • Computer vision • Autonomous systems • Blockchain/Web3 • Snowboarding • Sailing • Hockey • Track & Field